Sofia Residential Real Estate Market Analysis

Project Overview

Sofia's real estate market is characterized by substantial differences in property prices, rental rates, and housing affordability across the city's districts. This project analyzes apartment sale and rental prices, the impact of property characteristics on value, and the investment attractiveness of different areas using open data.

The study combines statistical analysis, geospatial visualization, and mortgage scenario modeling to provide a comprehensive view of Sofia's residential real estate market.

Business Problem

The analysis focuses on practical questions relevant to investors, buyers, real estate agencies, and market analysts:

- Which Sofia districts have the highest median price per square meter?
- Which districts offer the highest gross rental yield?
- How does housing affordability differ across districts?
- Do the construction period and number of rooms affect a property's relative value?
- Which apartment types are more attractive for buy-to-let investment?
- Can rental income cover mortgage payments under a standardized financing scenario?

Data Sources

The study uses open data from apartment sale and rental listings published on imot.bg. The data reflects the state of the real estate market as of June 2026.

Before the analysis, an automated collection process was used exclusively for publicly available listing data, without bypassing authentication mechanisms or access restrictions.

Housing affordability was assessed using data from Bulgaria's National Statistical Institute (NSI) on the average monthly salary of Sofia residents.

OpenStreetMap geospatial data obtained through the Overpass API was additionally used for spatial analysis.

Data Collection, Cleaning, and Exploratory Analysis

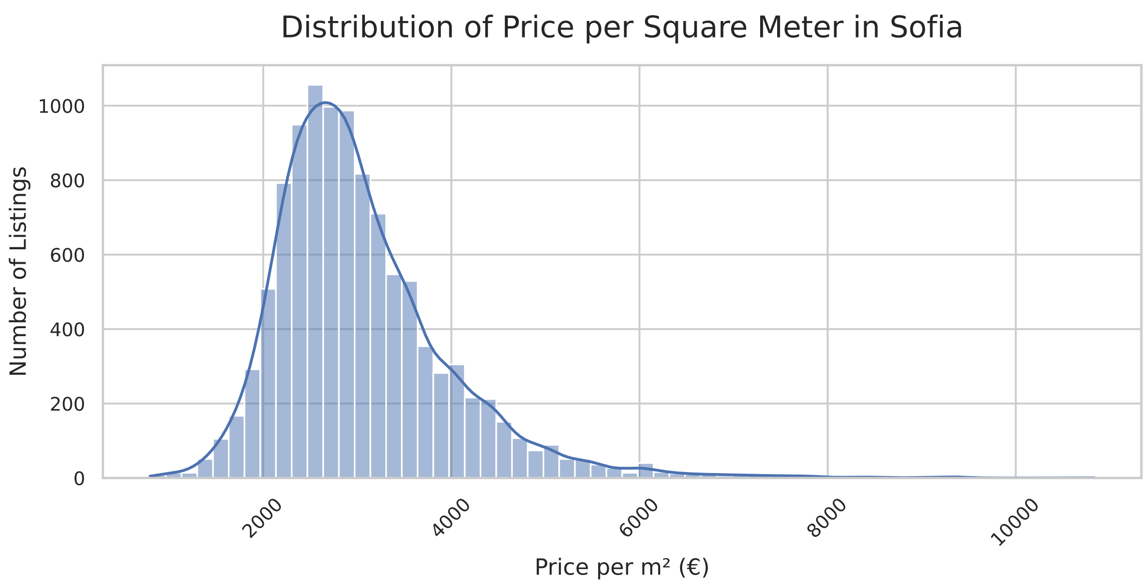
The study uses open data from apartment sale and rental listings published on imot.bg. Before the analysis, an automated collection process was used exclusively for publicly available listing data, without bypassing authentication mechanisms or access restrictions.

During data preparation:

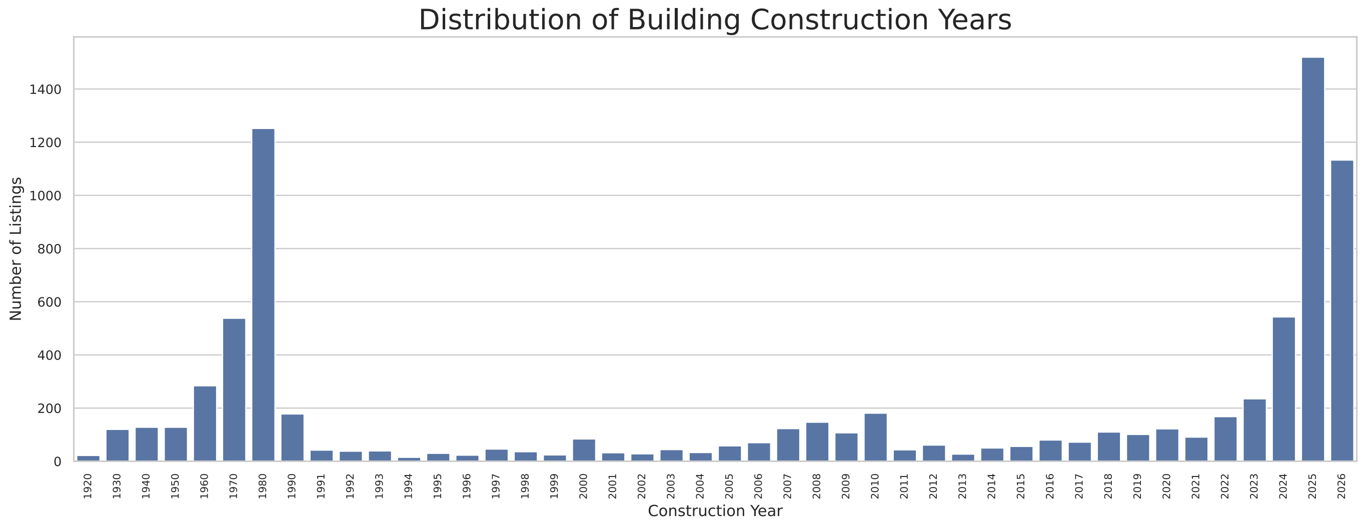
- missing and incomplete district geometries were corrected using geojson.io;
- data types were converted into appropriate formats;
- records with missing key characteristics were removed;
- properties still under construction were excluded;
- price-per-square-meter metrics were calculated for both sale and rental markets;
- district geometries for geospatial analysis were prepared using OpenStreetMap data obtained through the Overpass API. District names were matched and geometries were adjusted so that areas from imot.bg could be joined with spatial boundaries.

The three-sigma rule was applied to reduce the impact of extreme outliers.

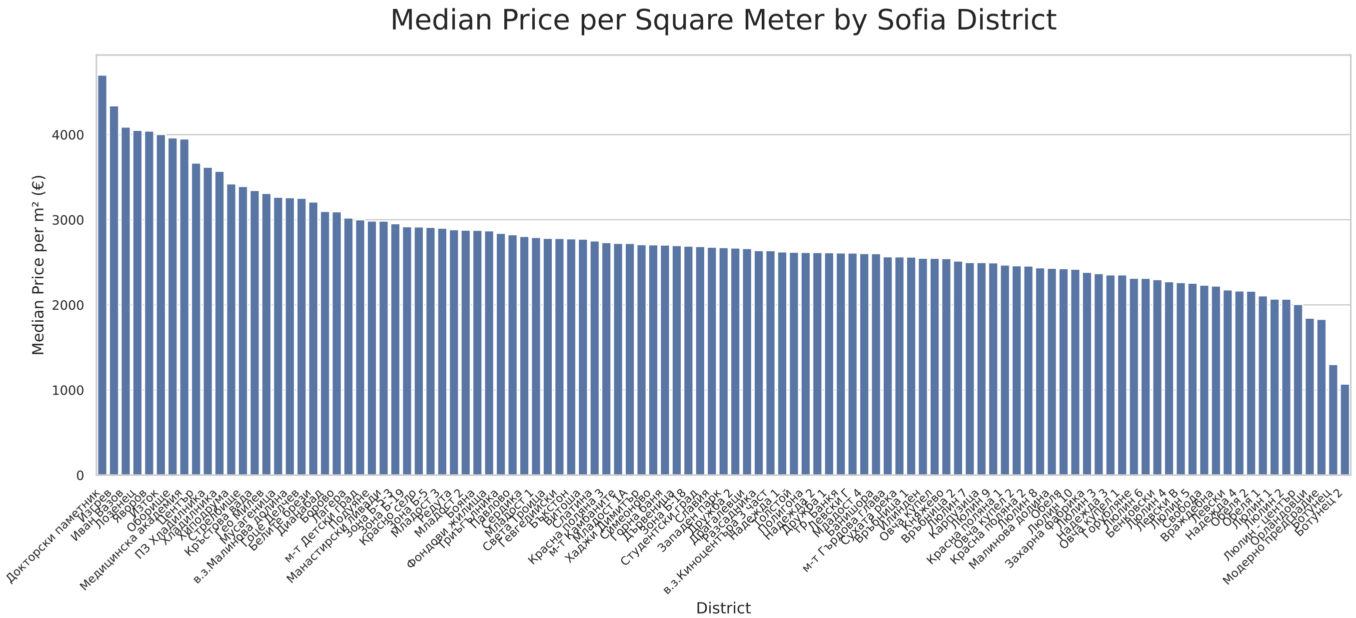
The price-per-square-meter distribution showed statistically significant deviations from normality. Therefore, non-parametric methods were used, and medians were selected as the main measure of central tendency.



In the sale market, most properties are priced between EUR 2,000 and EUR 4,000 per square meter, while a small number of premium listings create a long right tail.



The distribution by construction year shows two notable peaks in supply: properties built in the 1980s and new developments from 2024-2026. This provided the basis for a separate comparison of construction periods.



The median price per square meter varies substantially across districts. The most expensive areas include Doktorski Pametnik, Iztok, Lozenets, Oborishte, and Ivan Vazov. Among the most affordable are Botunets 2, Botunets, Orlandovtsi, Moderno Predgradie, and Lyulin.

Statistical Analysis

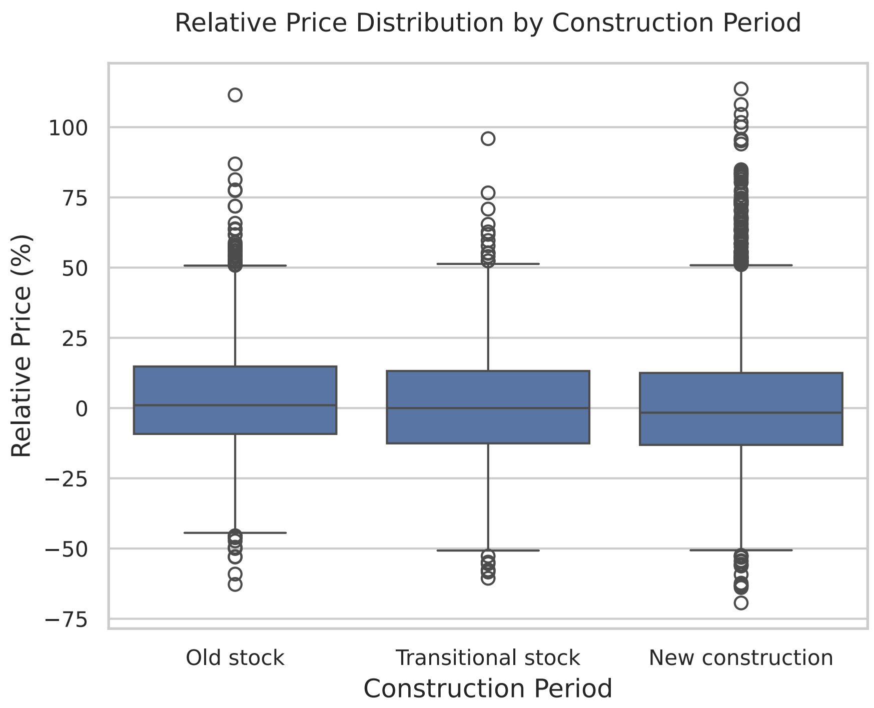
To control for location, a **relative** price per square meter was calculated against each district's median. This makes it possible to compare properties based on their characteristics rather than the general price level of the district.

Impact of Construction Period on Property Value

For the construction-period analysis, properties were divided into three groups:

- Old housing stock: 1900-1990.
- Transitional housing stock: 1990-2014.
- New developments: 2014-2026.

Differences between the groups were tested using the Kruskal-Wallis test, followed by Dunn's pairwise comparisons with Bonferroni correction.



Dunn's test indicates that the old housing stock forms a separate price group, while transitional properties and new developments have comparable relative price levels.

The median relative property value was:

Construction period	Relative price
Old housing stock	+1.02%
Transitional housing stock	0.00%
New developments	-1.64%

The old housing stock has the highest relative value, while transitional properties and new developments are valued less favorably by the market.

Change in Relative Property Value as the Number of Rooms Increases

A consistent relationship was identified between the number of rooms and relative property value.

The Kruskal-Wallis test and Dunn's multiple comparisons were used for the analysis.

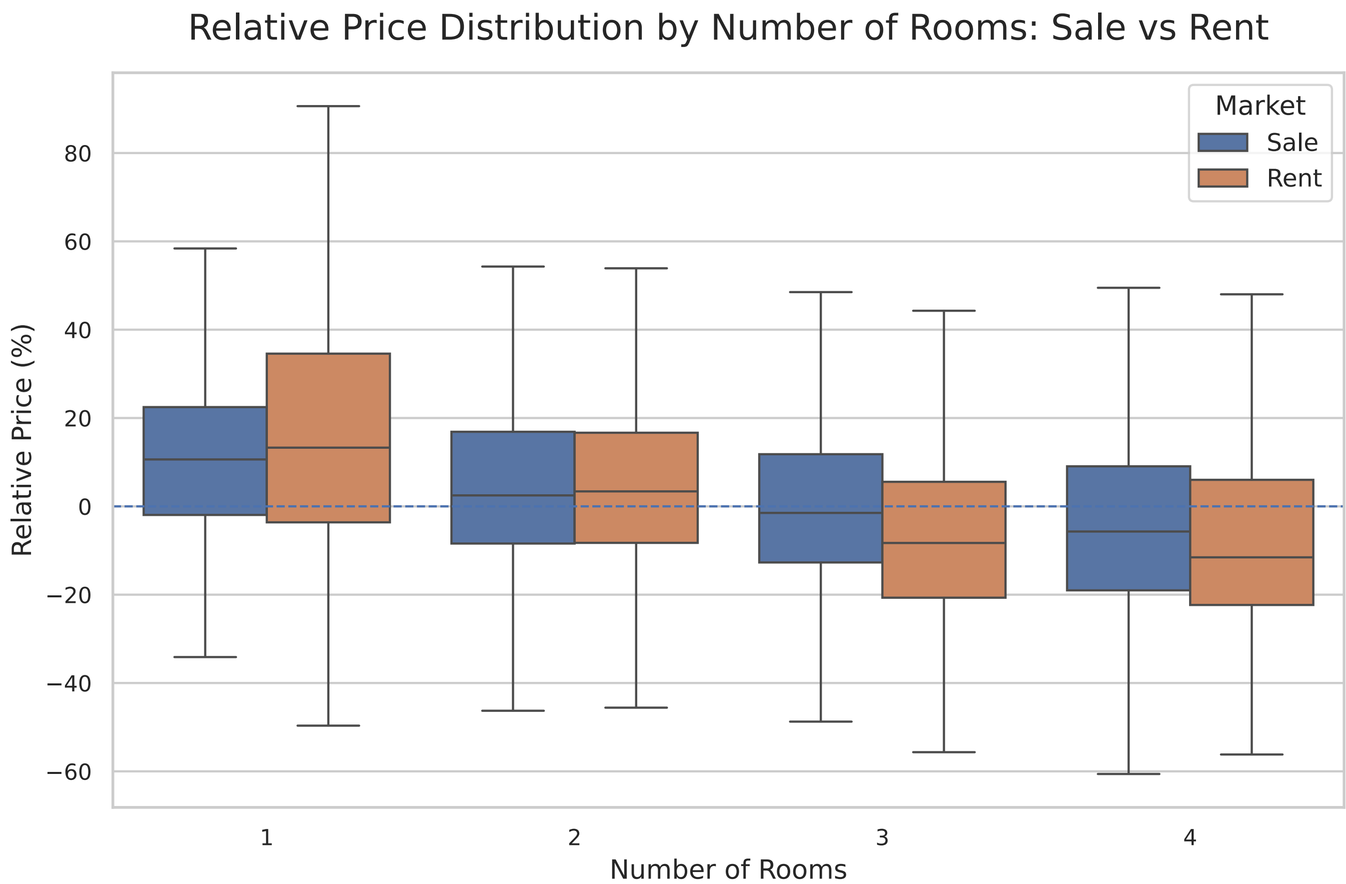
Main findings:

- one-room apartments are sold and rented at a premium relative to the typical

district price;
- two-room apartments also show a positive premium;
- three-room and four-room apartments are more often priced below the district median.

The results suggest stronger demand for compact apartments and a decline in price per square meter as property size increases.

- 1-room apartments: +10.62%.
- 2-room apartments: +2.48%.
- 3-room apartments: -1.49%.
- 4-room apartments: -5.70%.



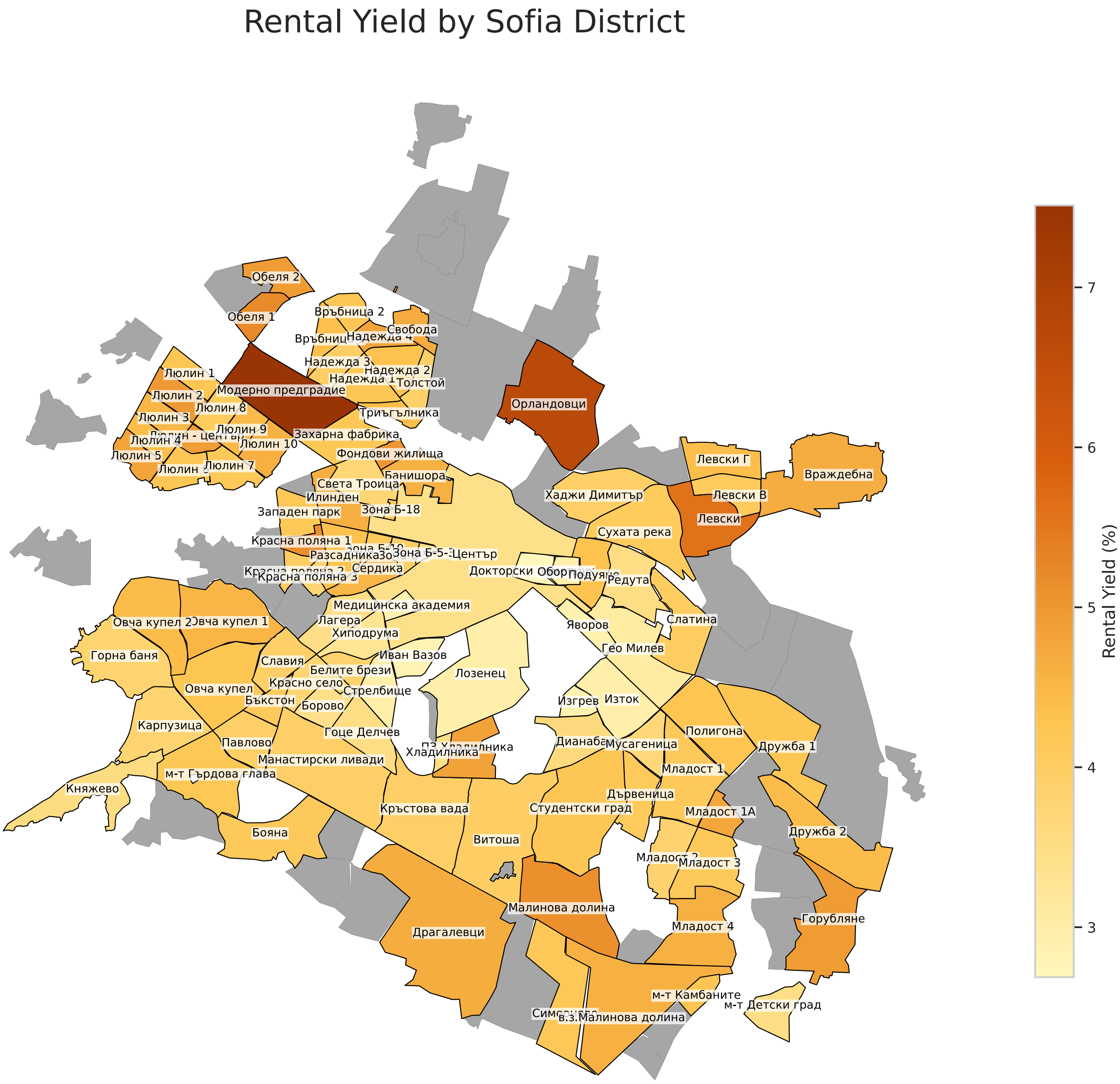
This pattern indicates stronger demand for compact apartments.

District Investment Attractiveness

The investment analysis calculates district-level indicators based on median sale and rental prices:

- `sale_price_m2`: median sale price per square meter.
- `rent_price_m2`: median monthly rent per square meter.
- `Yield_pct`: gross annual rental yield, %.
- `Payback_Period_years`: estimated payback period in years.
- `Months_to_buy_1m2`: housing affordability expressed as the number of average monthly

salaries required to purchase 1 m².

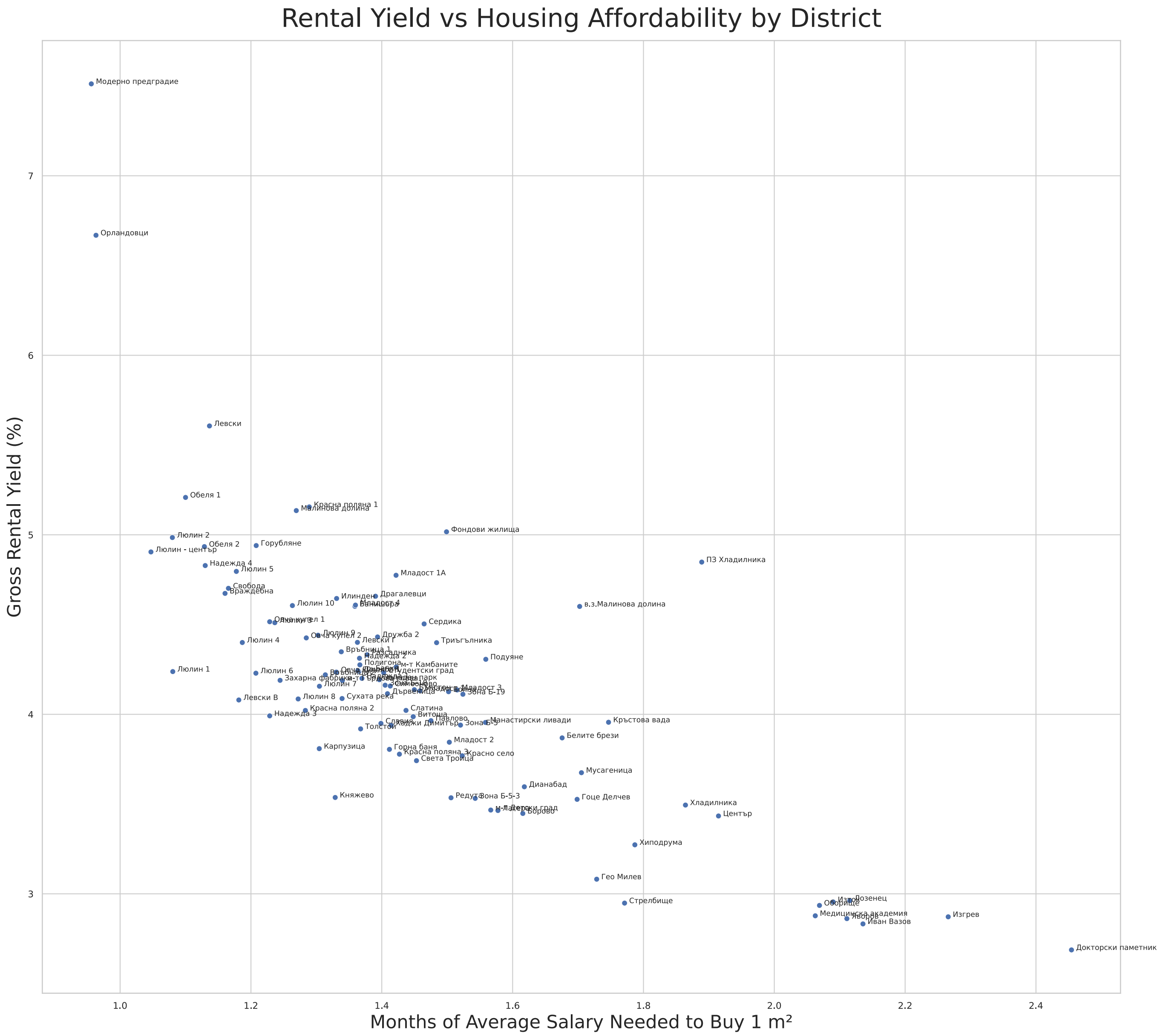


The rental-yield map highlights districts where the relationship between rent and purchase price is most favorable. These are not necessarily the most expensive districts. On the contrary, many high-yield areas are among the more affordable locations.

The highest gross rental yields in the cleaned dataset were:

District	Gross rental yield
Moderno Predgradie	7.51%
Orlandovtsi	6.67%
Levski	5.61%
Obelya 1	5.21%
Krasna Polyana 1	5.16%

The yield-and-affordability chart compares two business indicators at once: gross rental yield and the number of average monthly salaries required to purchase one square meter. Districts in the upper-left section combine higher yield with a lower entry cost.



Rental Yield Analysis by Apartment Type (number of rooms)

The Kruskal-Wallis test and Dunn's post-hoc test show that one-room and two-room apartments form a more attractive investment group. Three-room and four-room apartments are statistically similar to each other but deliver lower yields than compact properties.

The median gross rental yield was:

Apartment type	Median gross rental yield
1-room apartment	4.18%
2-room apartment	4.02%

Apartment type	Median gross rental yield
3-room apartment	3.46%
4-room apartment	3.41%

Therefore, one-room and two-room apartments are the most attractive property types for investment purchases.

Mortgage Self-Sufficiency

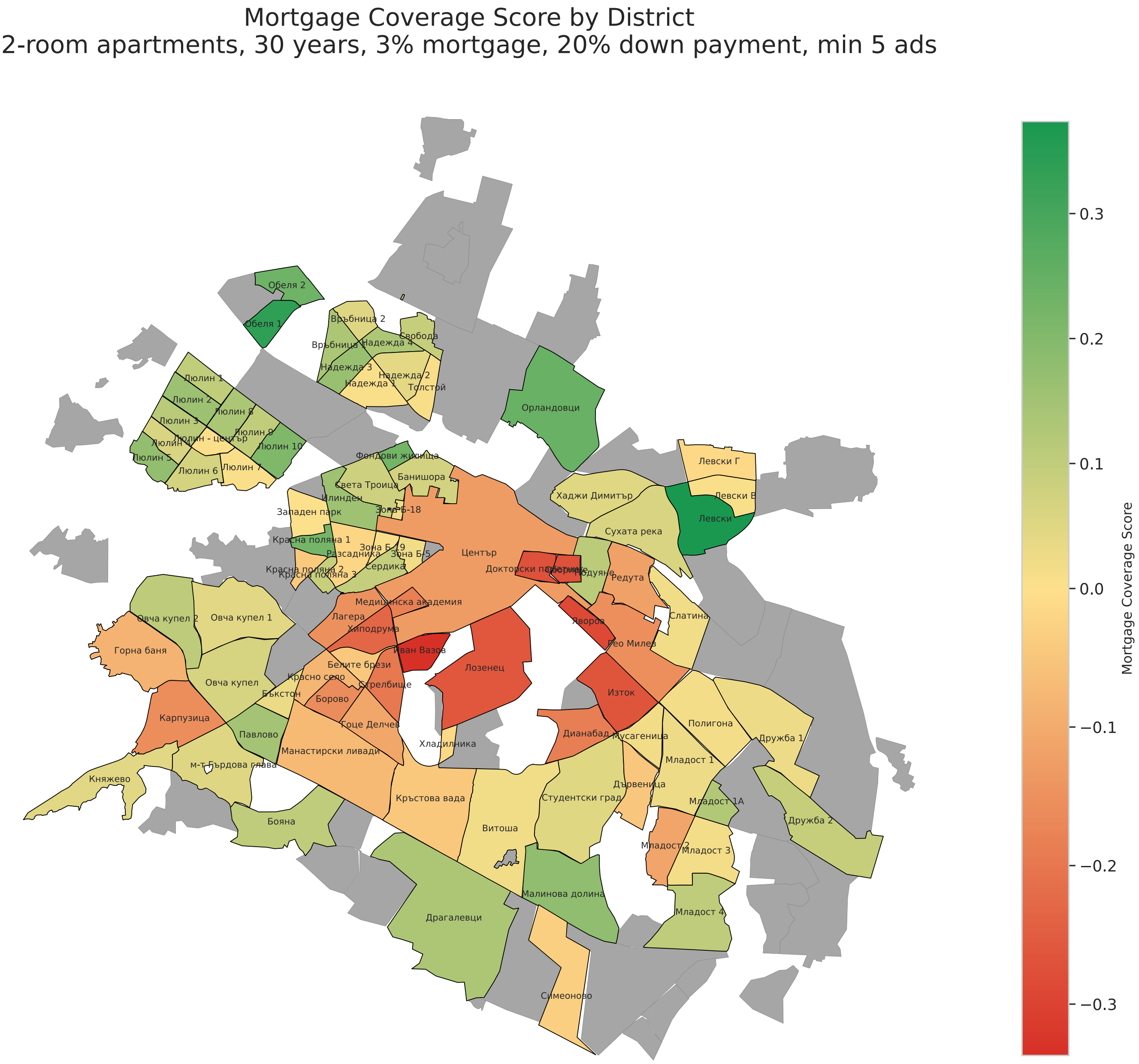
The mortgage scenario uses the following assumptions:

- loan term: 30 years;
- annual interest rate: 3%;
- down payment: 20%;
- standardized apartment area by number of rooms;
- at least 5 listings in a district for inclusion in the calculation.

For a typical two-room apartment, the strongest mortgage coverage is observed in Levski, Obelya 1, Orlandovtsi, Obelya 2, and Fondovi Zhilishta. The weakest results are found in premium districts such as Ivan Vazov, Yavorov, Oborishte, Doktorski Pametnik, and Iztok.

The Mortgage Coverage Score reflects the relationship between rental income and the mortgage payment:

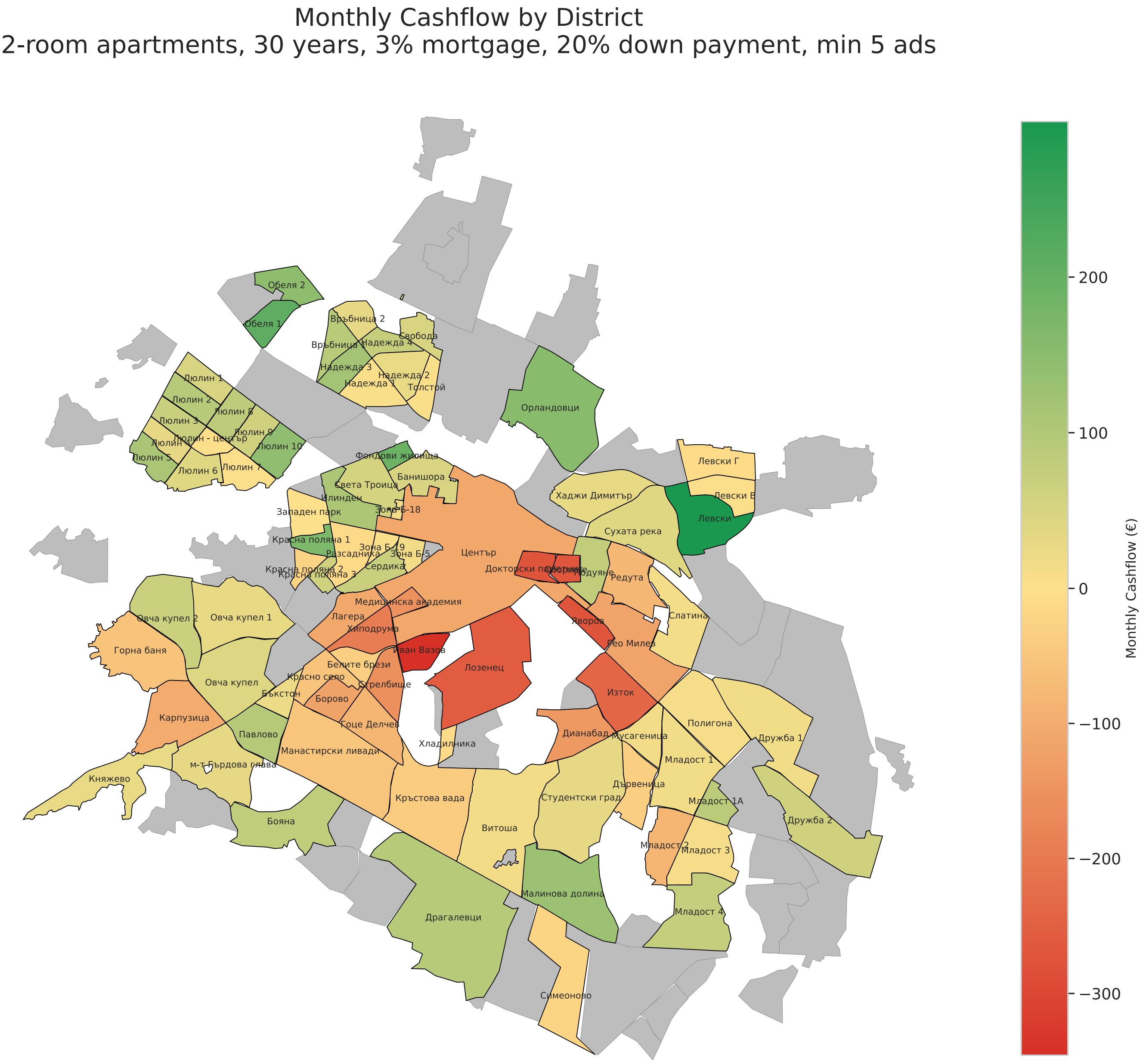
- 0 represents the break-even point;
- positive values mean that rental income exceeds the mortgage payment;
- negative values indicate that the owner must provide additional monthly funding.



Gray and white areas represent territories that were not included in the analysis: industrial zones, forests, water bodies, and districts with fewer than 5 available two-room apartment listings.

Monthly Cashflow was also calculated as the difference between rental income and the mortgage payment, expressed in euros.

- positive values indicate positive cash flow;
- negative values indicate a need for monthly out-of-pocket funding;
- green areas represent districts with the most favorable relationship between rental income and mortgage costs.



The mortgage maps add a financial layer to the analysis. They show that districts with similar gross rental yields can differ substantially in their ability to cover mortgage payments with rental income.

Key Findings

The study provides a comprehensive assessment of the investment attractiveness of Sofia's residential real estate market.

Main findings:

- Sofia's real estate market is highly heterogeneous across districts.
- Property value depends substantially on location.
- The old housing stock has a higher relative value than transitional properties and new developments.
- One-room and two-room apartments are the most attractive investment segment.

- Districts differ substantially in the ability of rental income to cover mortgage payments.
- Rental yield alone does not provide a complete assessment of investment attractiveness. Mortgage scenario analysis also considers whether rental income can cover the loan payment, providing a more realistic evaluation of investment performance.

Limitations

The project uses listing prices rather than completed transaction prices. Asking prices may differ from actual sale and rental prices.

The gross rental yield calculation does not account for taxes, vacancy, property management, repairs, insurance, transaction costs, or maintenance expenses.

The mortgage model uses standardized assumptions for the interest rate, loan term, down payment, and apartment area to compare districts. Actual lending conditions depend on the buyer and the bank.

Average salary is useful as a market-level affordability benchmark, but it does not reflect income distribution or the circumstances of individual households.

District boundaries and names had to be aligned between imot.bg and OpenStreetMap. Some spatial data required matching and manual correction.

Business Value

For investors, the analysis identifies districts where rental income has the greatest potential to support both the purchase economics and mortgage payments.

For real estate agencies, the project provides arguments for property positioning: which districts can be promoted based on affordability, which based on yield, and which apartment types are most relevant to rental investors.

For buyers, the analysis highlights the trade-off between a prestigious location, entry price, and potential rental income.

For recruiters and freelance clients, the project demonstrates a complete analytical workflow: public data collection, cleaning, feature engineering, statistical testing, geospatial analysis, business interpretation, and professional reporting.

Tools and Technologies

The project uses:

- Python.

- pandas and NumPy.
- SciPy and statsmodels.
- scikit-posthocs.
- Matplotlib and Seaborn.
- GeoPandas.
- Jupyter Notebook.
- GeoJSON.
- OpenStreetMap and Overpass API.
- Public listing data from imot.bg.